DAY 01 | HANDS-ON ACTIVITIES

Python Basics: Problem Statements

Eight problems from warm-up to stretch. Solve them in the Day 1 notebook or in a .py file in VS Code.

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How to work: read the problem, write the steps in plain English first, then write the code. Test with the sample run, then try your own inputs, including wrong ones.

Problem 1: Personal Intro Card

Level: Warm-up | **Concepts:** variables, print(), f-strings

Store your name, city, favourite food and years of experience in four variables. Print a neat introduction card using them.

Sample run

```
==============================
 Name       : Kenji
 City       : Pune
 Loves      : Masala dosa
 Experience : 3 years
==============================
```

Hints

- Use print("=" * 30) to draw a line.
- Use an f-string like f" Name : {name}".

Problem 2: Temperature Converter

Level: Easy | **Concepts:** input(), float(), arithmetic

Ask the user for a temperature in Celsius and print it in Fahrenheit. Formula: F = C x 9 / 5 + 32. Round to 1 decimal place.

Sample run

```
Enter temperature in Celsius: 36.6
36.6 C is 97.9 F
```

Hints

- input() returns text, so cast it with float().
- round(value, 1) rounds to one decimal place.

Problem 3: Even or Odd

Level: Easy | **Concepts:** int(), %, if/else

Ask for a whole number and tell the user whether it is even or odd.

Sample run

```
Enter a number: 17
17 is odd
```

Hints

- A number is even when number % 2 == 0.

Problem 4: Grade Calculator

Level: Medium | **Concepts:** if/elif/else, comparison operators

Ask for marks out of 100. Print the grade: 85 and above is A, 70 to 84 is B, 50 to 69 is C, below 50 is "Needs improvement". If the marks are below 0 or above 100, print "Invalid marks".

Sample run

```
Enter marks: 78
Grade: B
```

Hints

- Check the invalid range first.
- Order your elif checks from highest to lowest.

Problem 5: Multiplication Table

Level: Medium | **Concepts:** for loop, range()

Ask for a number and print its multiplication table from 1 to 10.

Sample run

```
Enter a number: 7
7 x 1 = 7
7 x 2 = 14
...
7 x 10 = 70
```

Hints

- range(1, 11) gives 1 to 10.

Problem 6: Simple Bill with Discount

Level: Medium | **Concepts:** float(), conditions, f-strings

Ask for the price of an item and the quantity. If the total is 2000 or more, apply a 10% discount. Print the subtotal, the discount and the final amount with 2 decimal places.

Sample run

```
Price: 450
Quantity: 5
Subtotal : 2250.00
Discount : 225.00
To pay   : 2025.00
```

Hints

- Use f"{value:.2f}" for two decimal places.
- Set discount = 0 first, then change it inside the if.

Problem 7: Number Guessing Game

Level: Challenging | **Concepts:** while loop, break, conditions

Store a secret number between 1 and 20. Let the user guess up to 5 times. After each wrong guess, say "Too high" or "Too low". If they guess correctly, congratulate them and stop. If they run out of attempts, reveal the number.

Sample run

```
Guess (1-20): 10
Too low
Guess (1-20): 15
Too high
Guess (1-20): 13
Correct! You got it in 3 attempts.
```

Hints

- Keep an attempts counter and increase it every round.
- Use break when the guess is correct.

Problem 8: Stretch: Pattern Printer

Level: Stretch | **Concepts:** nested loops

Ask for a number of rows and print a right-angled triangle of stars. Bonus: print it upside down as well.

Sample run

```
Rows: 4
*
* *
* * *
* * * *
```

Hints

- An outer loop handles rows, an inner loop prints stars.
- print("* ", end="") keeps output on the same line.

Ganbatte (頑張って): do your best